

CORNERSTONE PDK Monitor

User Manual

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TAPR Open Hardware Licence v1.0

Overview

The CORNERSTONE PDK Monitor is a free, single-file, browser-based dashboard for managing and sharing integrated photonics Process Design Kits (PDKs). It acts as a gateway between designers and fabrication foundries — giving the community a single place to discover, use, and contribute to open PDKs. No server, no installation: open `pdk-monitor.html` in any modern browser.

1. First-Time Use — Quick Start

1. Open `pdk-monitor.html` in Chrome, Firefox, or Safari. No installation required.
2. Accept the licence terms on first launch (TAPR Open Hardware Licence v1.0).
3. Explore the pre-loaded PDK — browse platform tabs, click BB cards, view files.
4. ****Switch to Admin mode**** to add/edit BBs. Open the HTML in a text editor and follow the comment at the top of the `<script>` block.
5. ****Connect to GitHub**** (optional) — click ☐ in the header, paste your PAT, enable pool fetching and publishing.
6. ****Configure the AI Agent**** (optional) — click  **Settings**, enter your API key, click ☐ Test connection.
7. ****Import your own data**** (optional) — click Import to load a `dashboard.json` and replace the default configuration.

2. Header Controls

Control	Function
 (refresh)	Hard reload — reloads from disk, bypassing the browser cache
Search box	Full-text search across BBs, platforms, contributors, descriptions
Status / Category dropdowns	Filter the BB card list

Control	Function
☐ GitHub	Configure GitHub PAT and user identity
☐ Insight	Open the AI Agent
☐ About	Version info, notes, editable links and logo
Import	Load a dashboard.json or ZIP backup
Backup (.json)	Export the full dashboard as a JSON backup
 Sync all	Replay all previously recorded GitHub fetch URLs

Storage indicator (in header): shows localStorage usage. Turns amber at 70%, red at 90%.

3. Platform Tabs

Each tab represents one fabrication platform. The tab shows platform name, material (Si / SiN / LN), BB count, and a completion progress bar.

- **Adding a platform:** Click **+ Add tab** (Admin only)
- **Reordering:** Drag tabs left or right
- **Platform settings:** Click the  icon to edit name, material, thickness, GitHub folder URL

4. Building Block (BB) Cards

Each card shows name, type icon, status pill, material badge, EDA tools, file count, contributor, and CDI maturity axes.

A small faded **selection checkbox** sits to the left of each card (Admin only). Tick it and use  **Move selected** to move multiple BBs to another platform at once. Click any card to open the BB Detail panel.

5. BB Detail View

Details

Edit name, description, type, category, status, contributor, CDI maturity levels, EDA tools, and planned release date.

Files

Upload or link GDS/OASIS, S-matrix (  , CSV), images, Python scripts, and documents.

Tests

Log characterisation results with date, contributor, pass/fail. Attach raw data files to each test entry.

AI

AI-generated BB summary (executive sentence + deep-dive, citing only measured values). Cached until you click  **Refresh**. The  **Ask AI** button pre-fills chat with this BB's context.

Parameters

Structured parameter table (width, gap, coupling length, etc.) editable per sub-entity or platform variant.

6. Data & File Management

Uploading files

Drag onto a BB card or use the upload button. Supported: GDS, OASIS, s2p, s4p, CSV, TSV, PNG, JPEG, PDF, PY, KPY, LYML, YAML, Markdown, and more.

GDS viewer

GDS files render inline with layer colours from the GDS layer map. Hover a shape for layer name. Edit via **Manage Lists** → **GDS Layers**.

S-matrix viewer

Upload a Touchstone (**.s2p**, **.s4p**, **.sNp**) or CSV file to plot $|S|$ vs wavelength/frequency.

Python script runner

Scripts tagged as Python run in-browser via Pyodide. Use ☐ **Run locally** for scripts needing native binaries or large datasets.

7. Multi-User Pool & GitHub

Setting up GitHub

8. Click ****☐** in the header
9. Paste a GitHub PAT with **repo** scope
10. Optionally enter your GitHub username / email
11. Tick ****Remember token**** to persist on this device

Fetch panel (section 2)

Button	What it does
☐ Fetch all enabled sources	Fetches the configured main repo and optionally all user slices
☐ Discover user slices	Scans the users/ folder for contributor dashboards
☐ Fetch my slice	Fetches only your own users/<id>/ slice
☐ Fetch user repos	One-click full pool assembly including all real files
Merge into my dashboard	Additively combines fetched data with local copy
☐ Replace whole dashboard	Overwrites local data with fetched content
☒ backup	Tick to auto-download a backup JSON before Replace

Tip: Untick ☒ backup if you run many fetch cycles and don't want a new file each time.

Publishing

Use ☐ **Publish** (Admin) to commit `dashboard.json` and all file changes to the configured GitHub repository.

8. AI Insight Agent

The AI Insight agent works with Claude, OpenAI, or any OpenAI-compatible endpoint (Ollama, LM Studio, etc.).

8.1 Configuration

Click ☐ **Insight** →  **Settings**.

Field	Notes
API endpoint	Default: <code>https://api.anthropic.com/v1/messages</code> . OpenAI: <code>https://api.openai.com/v1/chat/completions</code>
Model	<code>claude-sonnet-4-5</code> , <code>gpt-4o</code> , <code>llama3</code> , etc.
API key	Stored only in this browser's localStorage
Monthly token cap	Hard limit on tokens per calendar month
Egress mode	<code>metrics-only (safest)</code> · <code>metrics + extracts</code> · <code>full</code>
Prompt style	<code>Brief</code> · <code>Balanced</code> · <code>Deep</code>

8.2 BB Summariser

Open any BB → ☐ **AI tab** → ☐ **Summarise this BB**. Metrics extracted in-browser, only computed JSON sent to LLM. Cached. Click ☐ **Refresh** to regenerate. Also supports ☐ **Executive summary** for a whole platform or dashboard.

8.3 Multi-Turn Chat

Click ☐ **Insight** → ☐ **Chat with the agent**. Agent calls tools automatically:

Tool	What it reads
<code>list_platforms</code>	All platforms and BB counts
<code>list_bbs</code>	BBS filtered by platform, category, status, contributor
<code>get_bb</code>	Full detail for a specific BB
<code>get_bb_metrics</code>	Computed metrics for a BB
<code>search_bbs</code>	Full-text search
<code>summarise_bb</code>	Generates and caches an AI summary

8.4 GitHub File Fetcher

Inside the chat, expand ► **Data files in context** and paste a URL:

URL format	Example
Single file (blob)	<code>github.com/org/repo/blob/main/data.csv</code>
Folder	<code>github.com/org/repo/tree/main/measurements/</code>

URL format	Example
Multiple (newline or comma)	Paste several URLs at once

Click **Fetch** or press **Ctrl+Enter** / **Cmd+Enter**. Files appear as rows — click ☐ to toggle in/out of context. The file list scrolls when many files are loaded.

8.5 Analyse Files with AI

With files active (☐) , click ☐ **Analyse files with AI**. Client-side pre-analysis first (no tokens used): CSV stats (outliers, correlations), JSON schema, YAML keys. Pre-computed report then sent to LLM for insight bullets and anomaly flags.

8.6 Detachable Chat Window

Click ☐ **Detach** in the chat footer. Drag the title bar, resize from the bottom-right corner, minimise to a pill, or ☒ **Dock** back. Position and size are saved in localStorage.

9. Roles & Access Control

Role	Permissions
Admin	Full access: add/remove/edit all BBs, platforms, users, publish, manage lists
User-admin	Edit BBs in users/<id>; read-only elsewhere
Read-only	View all data, run scripts, download files; cannot edit or publish

The default role is Read-only. To switch to Admin, see the comment at the top of the `<script>` block in the HTML file.

10. Import / Export & Backup

- **Backup (.json)** — Downloads full dashboard. Use to back up data or commit to GitHub
- **Import** — Loads a `dashboard.json` or ZIP, replacing the current model. Auto-backup first
- **Export BB folder (zip)** — BB → Files tab → Download folder: ZIP with all BB files and `README.md`
- **Run locally (zip)** — Python scripts: self-contained folder with script, data files, and `run.sh`

11. Keyboard Shortcuts

Shortcut	Action
/	Focus search box
Escape	Close modal / detail panel
Ctrl+Enter / Cmd+Enter	Send AI chat message; fetch files in URL input
Ctrl+S / Cmd+S	(Script editor) Save script

12. Troubleshooting

Dashboard shows nothing / nothing is clickable

Hard reload: Cmd+Shift+R (Mac) or Ctrl+Shift+R (Windows). If that fails, close the tab and re-open the HTML file directly.

AI agent returns 'No API key'

Open   Settings. Enter your API key and click ☐ Test connection to verify.

API error 429 (rate limit)

Your organisation hit the per-minute token cap. Collapse the file panel (click ►) before chatting to reduce tokens per message. Or switch to a faster model (Haiku).

GitHub fetch returns 403

Hit the 60 req/hr unauthenticated limit. Add your GitHub PAT in ☐ settings (5000 req/hr).

File fetcher loads fewer files than expected

Files > 2 MB are skipped. Binary files (GDS, images, compiled) are excluded by design.

Storage indicator is red

Near the localStorage limit. Export JSON, then use ☐ **Clear AI cache** in AI Settings and delete old backups.

Dashboard shows stale data after a pool fetch

Click ↺ in the header to hard reload.

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